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An LLM-Based Resume Intelligence System for Recruiters and Job Seekers

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Abstract

The increasing volume of job applications in modern recruitment workflows has revealed clear limitations in traditional resume screening methods, which largely depend on keyword matching and basic filtering techniques. These approaches often struggle to capture contextual skills, lack deeper semantic understanding, and provide little to no feedback for candidates. Additionally, most existing systems are designed primarily for recruiter-side filtering, offering minimal assistance to job seekers.

This paper presents an LLM-Based Resume Intelligence System that introduces a dual-mode framework to support both recruiters and job seekers within a unified architecture. The system employs Large Language Models to perform context-aware resume parsing, structured information extraction, and semantic skill interpretation. To enable scalable similarity analysis and efficient retrieval, resumes and job descriptions are transformed into vector embeddings and indexed using FAISS. In recruiter mode, the system facilitates automated bulk resume screening, role-fit evaluation, and candidate ranking based on semantic relevance. In job seeker mode, it provides resume insights, identifies potential skill gaps, and generates targeted improvement guidance.

The proposed design emphasizes scalability, explainability, and practical deployment feasibility by avoiding computationally expensive multi-model ensembles. Preliminary evaluations using ranking-oriented metrics, including Precision@K and Mean Reciprocal Rank (MRR), indicate consistent retrieval behaviour and reliable role-fit assessment. The results suggest that integrating LLM-driven semantic analysis with vector search techniques can enhance recruitment efficiency while providing meaningful decision support for both stakeholders.

Keywords: Large Language Models (LLMs), Resume Intelligence System, Semantic Similarity, FAISS Vector Search, Resume Parsing, Skill Gap Analysis, Candidate Ranking, Embedding-Based Retrieval.

1. Introduction

The fast-paced digital transformation of recruitment procedures has reshaped modern hiring practices. In fact, the growth of online job portals and professional networking platforms has resulted in employers receiving a substantial number of resumes for each vacancy. Although such a development has expanded the pool of candidates, it has simultaneously brought about difficulties for hiring managers to comprehend and evaluate applications in a limited period. Therefore, there is a compelling need for automated recruitment systems.

Majority of current Applicant Tracking Systems (ATSs) depend on simple rule-based and keyword matching methods. These approaches can be considered as basic automation

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solutions only. They are devoid of any contextual understanding and fail to handle different resume layouts properly. Besides, they cannot interpret experience accurately. Hence, this leads to the situation where worthy candidates are skipped, and at the same time, less relevant profiles keep passing through screening stages.

Current software is recruiter-oriented, which is another significant shortcoming. It means that job seekers do not get much analytical feedback from these systems. Without getting feedback on the quality of the resume, skills that are missing, or the alignment with the role requirements, candidates are limited in their capacity to enhance their profiles and make adjustments as per the needs of the industry.

Large Language Models (LLMs) have been recently improved to a great extent in semantic understanding, contextual reasoning, and information extraction. By combining these breakthroughs with embedding-based representations and vector similarity search infrastructures, the technologies become capable of scalable and accurate resume-job matching.

On the other hand, this article is about a dual-mode resume intelligence system that is beneficial to both recruiters and job seekers. The incorporation of LLM-based semantic analysis and vector similarity search in this system leads to increased accuracy in screening, making recruitment workflows both efficient and adaptive.

2. Literature Survey

Initial solutions for automated hiring processes mostly depended on keyword matching and rule-based parsing which increased the speed of processing but these systems lacked semantic understanding and had difficulties with diverse resume formats.

Recent studies in automated hiring processes have moved greatly into resume parsing, candidate screening, and job-role matching by employing NER (Named Entity Recognition) and other deep learning techniques based on transformers have significantly improved, not only, the extraction of structured information but also contextual understanding. However, these methods have a common problem that they need huge annotated data and high computational power. Besides, the majority of the current recruitment systems are solely for the recruiter's side of the process and they hardly ever provide analytical feedback for the job seekers.

Semantic embedding representations along with similarity searches, for instance, FAISS, have shown a very good level of performance in resume-job matching at scale. Therefore, the present work suggests the integration of LLM-based semantic parsing with embedding-based retrieval and introduces a dual-mode framework supporting both recruiter screening and job seeker guidance.

Existing System	Limitation
Keyword Matching	Lacks semantic understanding and contextual interpretation
Rule-based Parsing	Poor generalization across diverse resume formats
Machine Learning Classifiers	Requires large labeled datasets and extensive training
Traditional ATS Platforms	Provides no meaningful feedback to job seekers

Table 1: Comparison of Existing Systems and Identified Limitations

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3. Proposed System Architecture

The proposed system employs a modular and scalable structure to accommodate recruiter and job seeker features in single framework.

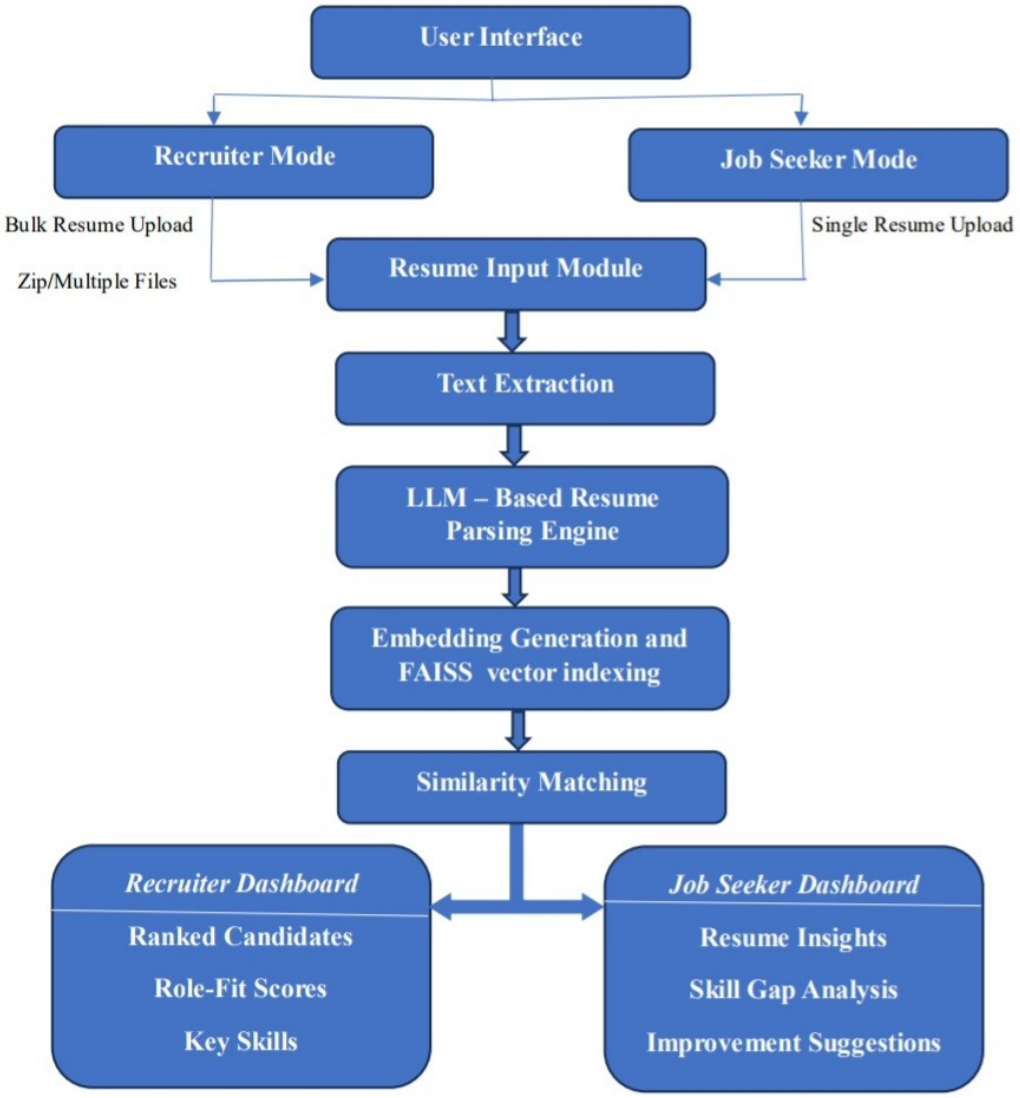


Figure 1 : System Architecture of the Proposed LLM-Based Resume Intelligence System

3.1 Unified User Interface

It is a dual-mode interface that can be switched to recruiter mode for bulk resume upload, or job seeker mode for single resume evaluation.

3.2 Resume Input and Text Extraction Module

This module receives the resumes either as PDF or DOCX files, extracts clean text, removes noise, and normalizes the content for the subsequent steps.

3.3 LLM-Based Resume Parsing Engine

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It conducts an in-depth semantic analysis of the text to identify the corresponding fields like education, experience, skills, certifications, and projects.

3.4 Embedding Generation Module

This module transforms the content of the parsed resumes into compact vector-based representations with the help of already-trained embedding models, thus capturing semantic relationships.

3.5 FAISS Vector Indexing Module

It keeps embeddings indexed to be used in a quick and scalable similarity search through a large collection of resumes.

3.6 Similarity Matching and Ranking Module

It is responsible for calculating the cosine similarity between resumes and job descriptions, thus providing ranked candidate lists and the role-fit scoring of each candidate.

3.7 Output Visualization and Feedback Module

It uses dashboards to share the result and tools with recruiters to allow them to see ranked profiles and acquire insights; similarly, for job seekers, it offers resume analysis, skill gaps, and suggestions to make improvements.

This architecture combines the workflows of both the recruiter and the job seeker and provides an intelligent, scalable, and cost-effective recruitment platform.

4. Methodology

This section presents the methodological approach of the proposed LLM-driven resume intelligence system. The architecture of the whole system is well laid-out, it firstly takes a resume for input and finally it outputs the visualization of results and feedback generation. The two types of users the system serves are recruiters and job seekers, thus it delivers fast resume parsing, semantic analysis, similarity matching, and decision-making assistance.

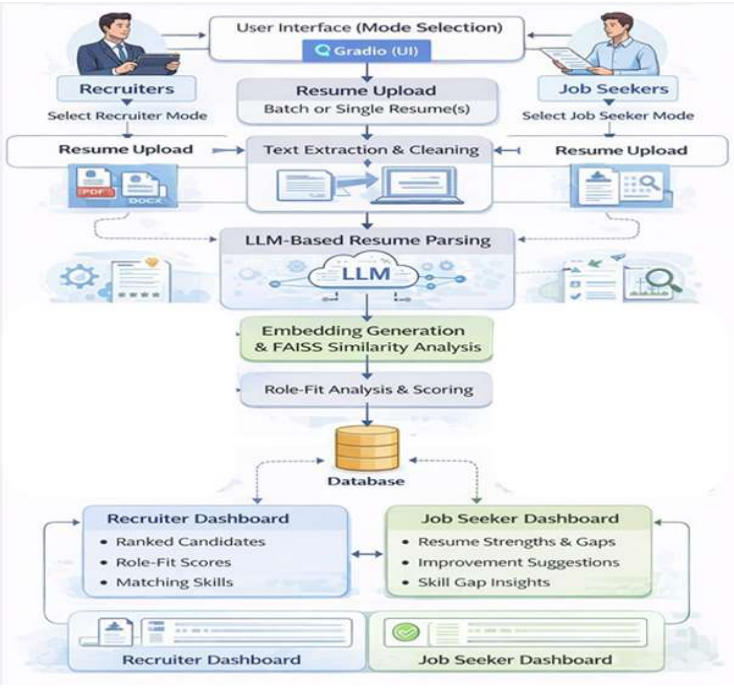


Figure 2 Methodology of Proposed System

4.1 Resume Upload and Input Handling

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Recruiting mode allows for uploading a batch of resumes at once in zip files or multiple-file formats to quickly filter through a large number of applicants. On the other hand, the job seeker mode is meant for one-person, one-resume upload for receiving highly targeted advice. The system can process commonly used formats such as PDF and DOCX which are user-friendly and compatible.

4.2 Text Extraction and Pre-processing

Text extraction from resumes starts automatically once the documents have been uploaded and the content is then extracted. The extracted text is then pre-processed by getting rid of the noise, normalizing the white spaces, and cleaning the text to keep it free from the inconsistencies and make it accurate for semantic analysis.

4.3 LLM-Based Resume Parsing and Information Extraction

After cleaning, the text is fed into an LLM-based parser that identifies the information tied to the various resume sections e.g., personal, education, work experience, skills, certifications, and awards. The LLM-powered parser goes beyond the capability of rule-based solutions as it delivers the great understanding of the content in the face of various resume layouts and heterogeneous writing styles.

4.4 Embedding Generation

The information that has been parsed is encoded into dense vector representations by using pre-trained sentence embedding techniques. The embeddings are capable of reflecting the semantic proximity between different segments of resumes and job descriptions, which leads to enhanced similarity measurement that goes beyond the surface-level simple keyword matching.

4.5 FAISS-Based Vector Indexing and Similarity Matching

In order to carry out efficient and scalable retrieval, the embeddings generated are first indexed with the help of Facebook AI Similarity Search (FAISS). Later, the similarity is determined between two sets of vector representations, i.e., the profile of the candidates and the requirements of the job, by making use of cosine measure. Thus, the desired candidates can be effectively filtered out in no time.

4.6 Role-Fit Scoring and Candidate Ranking

From the similarity values, a role-fit measure is derived that is meant to reflect the level of candidate suitability. Resumes are then sorted based on these scores so that the shortlisting can be automated and the time investment can be minimized while still being fair and consistent.

4.7 Output Visualization and Feedback Generation

The system, of course, in the case of the recruiter, it offers a variety of functionalities such as a sortable list of candidates, their role-fit scores, and skills extracted through a highly interactive interface. However, when viewed from the job seeker perspective, the system generates a resume critique, indicates the missing skills, and offers the set of guidelines to improve one's resume that is geared not only towards career growth but also towards re-skilling.

5. Experimental Results and Performance Evaluation

5.1 Experimental Setup

The proposed system was evaluated using a dataset consisting of real-world resumes and job descriptions collected from publicly available sources. The resumes were processed through the LLM-based parsing pipeline, and their corresponding semantic embeddings were

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generated. These embeddings were indexed using FAISS to enable efficient similarity-based retrieval.

Multiple job descriptions were tested against the resume dataset, and ranked candidate lists were generated using cosine similarity. The experimental evaluation mainly focuses on recruiter mode, as it represents the primary operational workflow of the system

5.2 Recruiter Mode Results

	rank	name	title	skills	similarity
20	1	Aryaman Rao	Information Technology Specialist	Windows, Unix, Linux, VMWare, Wireshark, Snort...	0.759405
95	2	Keshav Kumar	Information Technology Specialist	COMMUNICATION, Cloud Environments Design & Imp...	0.754810
98	3	Sravani Kumar	Information Technology Specialist	Desktops, Ethernet cables, Cisco routers, Vide...	0.747421
61	4	Shivanya Rao	Information Technology Specialist	INFOSEC/NETOPS, CCRI, ACAS, SCADA, C4ISR, IT, ...	0.744716
0	5	Riya Verma	Information Technology Specialist	Remote access support, Systems analysis, Preve...	0.731390
58	6	Riya Malhotra	Information Technology Specialist	account management, cables, cabling, Help Desk...	0.725615
30	7	Eesha Gupta	Information Technology Specialist	Hardware updates, Process improvement implemen...	0.721638
36	8	Aman Sharma	Information Technology Specialist	active directory, antivirus, C, cable, cat5, c...	0.719313
3	9	Dhanush Malhotra	Information Technology Specialist	Computers, trouble shooting, Microsoft Access,...	0.711628
105	10	Abhinav Mehta	Manager - Information Technology and Building ...	System administration, Windows Server 2003, Wi...	0.703118
71	11	Jahnavi Verma	Information Technology Specialist	Active Directory, administrative support, Adob...	0.692264
39	12	Kavya Reddy	Information Technology Specialist	Accounting, Active Directory, anti-virus, Appl...	0.690074
90	13	Keshav Iyer	Information Technology Specialist	Army, Cisco, counseling, customer assistance, ...	0.687171
112	14	Dhanush Patel	Information Technology Specialist	Procurement Clerk, Purchasing Agent, Computer ...	0.678504
7	15	Priya Malhotra	Information Technology Provisioning Technician	A Certification, Active Directory, AD, Anti-Vi...	0.676697
22	16	Varun Patel	Information Technology Specialist	A+, Active Directory, agile, Avaya, call cente...	0.676041
4	17	Pavani Mehta	Information Technology Specialist	Active Directory, Adobe, Antivirus, Backup Exe...	0.670333
28	18	Mrudula Mehta	IT Manager	Antivirus, Backup Exec, Backup, Cisco Router, ...	0.669181
114	19	Karthik Mehta	Director of Information Technology	Accounting, Analyst, business processes, Chang...	0.666580
32	20	Pavithra Kumar	Information Technology Specialist	issue escalation, reporting and analysis, qual...	0.664226

For a sample job description related to the role of Information Technology Specialist, the system retrieved the top-20 ranked candidates along with their similarity scores.

The similarity values ranged between 0.66 and 0.76, indicating strong semantic alignment between resumes and job requirements. The ranking results demonstrate the effectiveness of the system in identifying relevant candidates efficiently and accurately.

5.3 Evaluation Metrics

Ranking-based evaluation metrics such as Precision@K and Mean Reciprocal Rank (MRR) were used to quantitatively measure retrieval performance, as depicted in Figures 3 and 4.

```
for k in [3,5,10]:
    print(f"Precision@{k}: {precision_at_k(ground_truth, k):.3f}")
```

```
Precision@3: 1.000
Precision@5: 1.000
Precision@10: 0.800
```

Figure 3 : Precision@K at different cutoff values.

```
mrr = mean_reciprocal_rank(ground_truth)
print("MRR:", round(mrr,3))
```

```
MRR: 1.0
```

Figure 4 : Mean Reciprocal Rank (MRR).

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These results indicate that the most relevant candidates consistently appear within the top-ranked positions, validating the effectiveness of the embedding-based semantic matching approach.

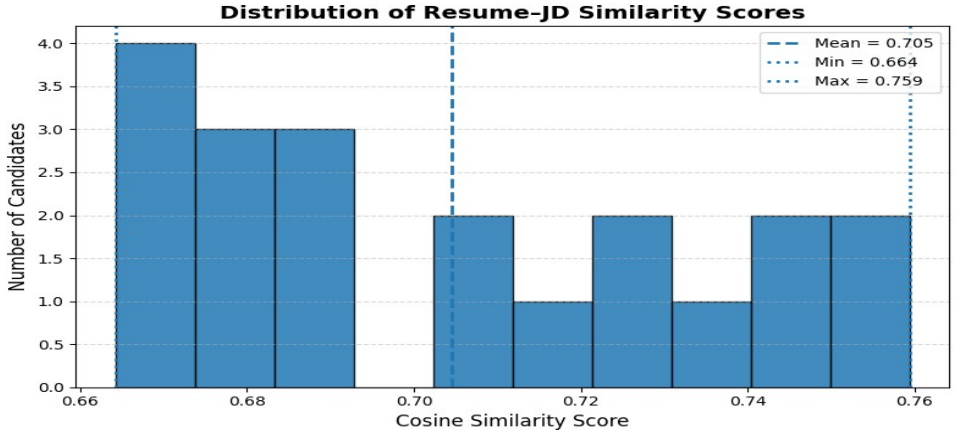


Figure 5: Distribution of Resume–JD Cosine Similarity Scores

Figure 5 shows the distribution of cosine similarity scores among the shortlisted resumes. The scores go from 0.664 to 0.759 with a mean of 0.705. This means that the selection of the candidates is stable and consistent with each other, in terms of their fit for the role.

5.4 Discussion on Job Seeker Mode

While the current experimental evaluation primarily emphasizes recruiter-side functionalities such as resume ranking and candidate matching, the job seeker mode represents an equally significant component of the proposed system. This module is designed to provide resume analysis, personalized feedback, identification of skill gaps, and targeted improvement suggestions using LLM-driven insights.

Due to ongoing implementation and optimization, extensive experimental validation of the job seeker mode is planned as part of future system enhancement. Preliminary outputs indicate promising potential in delivering actionable recommendations to job seekers, thereby improving resume quality and job-fit alignment.

6. Future Scope

Comprehensively, the LLM-based resume intelligence system can be significantly uplifted with an array of features covered under automation, intelligence, and usability. The gist of these proposed enhancements pivots around accomplishing end-to-end recruitment automation as well as equipping job seekers with superior career guidance.

6.1 End-to-End Automated Recruitment Workflow

At present, the system is limited to semantic resume screening and candidate ranking. Later, this system can be refurbished to handle completely automated recruitment workflows by fetching resumes and candidate details through platforms integrated with Google Forms, for example. The gathered data can be instantly and automatically utilized, examined, and graded using semantic similarity models.

Furthermore, according to the ranking results, the system could send customized automated emails to the shortlisted candidates about next steps of the hiring process. This automation chain will drastically cut down the manual work, shorten the recruiting time, and make a hiring process that is entirely self-operating possible.

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6.2 Intelligent Career Guidance for Job Seekers

While assisting job seekers, the platform may move far away from merely judging the resumes and spotting skill deficits, to actually issuing the personalized learning roadmaps as well. In view of the weaknesses uncovered, the system can propose appropriate educational materials, certification courses, and well-organized learning tracks to enhance skill development and thus, raise the candidates employability standards.

7. Conclusion

This paper presented an **LLM-Based Resume Intelligence System** designed to support both recruiters and job seekers through a unified dual-mode framework. By integrating context-aware resume parsing, semantic embedding generation, and FAISS-based similarity matching, the system effectively addresses the limitations of traditional keyword-based resume screening techniques.

Experimental evaluation on recruiter mode demonstrates reliable candidate ranking performance with high Precision@K and Mean Reciprocal Rank values, validating the effectiveness of semantic similarity-based retrieval. The proposed framework significantly reduces manual screening effort, enhances role-fit assessment, and improves recruitment efficiency.

Overall, the developed system provides a scalable, intelligent, and practical solution for modern recruitment workflows, with strong potential for future enhancements and real-world deployment.

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